Machine Intelligence 1

Course Organization

Lecture (Prof. Klaus Obermayer)
Fri 08:15 - 09:45 — MA 004
→ Lectures begin 17 Oct 2025.

Tutorials (you only need one)
Thu 08:15 - 09:45 — MA 004 (All study programs)
Thu 11:15 - 12:45 — BCCN lecture hall (HU campus - Computational Neurosci.)
→ Tutorials begin starting from 23 Oct 2025.

WS 25/26

Course material on the ISIS platform

Access:

- Subscribe with your TUB account.
- temporary guest access for non-TU students

There, you will find:

- lecture slides and literature references
- recorded videos and notes in case you missed a lecture or tutorial
- weekly exercise sheets, solutions, and exam results
- announcements & discussion forum

» Frequently check for updates «

For your weekly schedule

every Wednesday:

- new exercise sheet

every Thursday:

- tutorials
- solution for the exercise sheet of the previous week

every Friday:

- lecture

also:

- recorded videos in case you missed a lecture or tutorial

The Exercise sheets

- weekly exercises: programming[1] & theoretical problems
- programming resources, *Jupyter notebook* example/introduction available on ISIS
- designed to be solved in groups of 3-4 (reflected in the workload).
- discussion via ISIS is highly encouraged
- The solutions:
  - are posted 1 week later,
  - contain key results from the questions
  - The solution is not clear?
    - Ask during your tutorial session.
    - Post questions on the ISIS discussion forum.
- Hints to solve:
  - pseudocode usually available on the lecture (& tutorial) slides
  - not sure, if correct? $\rightarrow$ use off-shelf implementations as a baseline
  - google with words like "numpy" (if you use python) to get analytical approaches for implementation

[1] Your choice of programming language: Python(recommended), R, Julia or Matlab

The Tutorial

- Discussion of the solutions from the previous exercise sheet.
- Necessary theory for current exercise sheet
- Examples and discussion of lecture topics.

Earning credits: Modules

For MSc Computational Neuroscience:

- Machine Intelligence I and II are examined together (12 ECTS credits).
- exam eligibility criteria: $\geq$ 60% in 10 / 13 sheets
- oral exam

For all other study programs:

- only Machine Intelligence I is examined (6 ECTS credits).
- automatic eligibility for the exam
- Excercise sheets will not be graded. No handing-in of your solutions. Solving them is highly recommended.
- written exam

Final Grade

- The module grade is determined exclusively by the exam.

MSc Computational Neuroscience

1. Assemble into groups of **3-4 people**.
2. Send the list with group members to the `tutors` ✉ (use template on ISIS).
3. Solve sheets as a group, same group for the rest of the semester - if you want to switch it up, you have to notify me!
4. Submit your group's solution via ISIS by Wednesday evening.
5. Usage of off-shelf implementations is prohibited unless otherwise indicated on the sheet.

   For each sheet one zip file from each group containing:

   a. Jupyter notebook with source code and results for programming exercises
   b. PDF file with results:
      - filename: must contain sheet-number and group name
      - sheet: must contain names of group members
      - form: scanned hand-written solutions
      - form: additionally, PDF of Jupyter notebook & Jupyter Notebook.
6. Present your solution to the other groups (mandatory).
7. Presentation schedule will be announced as soon as groups are formed.

Oral Exam: Procedure for MSc Computational Neuroscience

- Eligibility criteria are met for MI I and II (see previous slide 6)
- Registration (details in the summer term).

Written Exam 1: Procedure for all other study programs

Important dates for the exam 1 on
17 Feb 2026 at 11:00 in H 0105
Registration
opens 17 Jan 2026 – 08:00 h
and closes 10 Feb 2026 – 10:30 h
Cancellation possible before
14 Feb 2026 – 10:30 h

Registration (deadline 10 Feb 2026 – 10:30 h):

- *TU Berlin students* register for the exam via Moses (MTS) online.
- *Erasmus students*: You can register via MOSES (MTS) as the others as well.
- *Visiting students that can not register via MOSES* register by filling out `this form` and sending it to our NI secretary (`camilla.groiss@tu-berlin.de` ✉) before 10 Feb 2026 – 10:30 h.
- Complete all of the above before 10 Feb 2026 – 10:30 h. Allow for time to get an extra signature/confirmation or other formalities if necessary.

Written Exam 2

- For people who:
  - did not pass their first attempt,

    OR

  - registered for a prior exam but did not attend it,

    OR

  - canceled their registration for a prior exam,

    OR

  - did not register for the first exam.
- **Important**: Registrations do not carry over.
  Every exam requires its own registration that is separate from registrations for any prior exams.
- In all cases, a new registration is required.

Written Exam 2: Procedure (separate registration required)

Important dates for exam 2 on
09 Apr 2026 at 08:00 in H 0105
Registration
opens 09 Mar 2026 – 08:00 h
and closes 02 Apr 2026 – 07:30 h
Cancellation possible before
06 Apr 2026 – 07:30 h

Registration (deadline 02 Apr 2026 – 07:30 h):

- Follow the same procedure that applies to you for *Written Exam 1* (watch out for different dates and deadlines).
- Allow for time to get an extra signature/confirmation or other formalities.
- Your registration needs to be completed before 02 Apr 2026 – 07:30 h.
- registration related questions?
  ⇒ Only Ms Groiß ✉ can answer them.

Withdrawing from the exam / Canceling your exam registration

- Possible until 3 days before the exam, after that[2], only possible with medical "Attest"[3].
  - ⇒ withdrawing from written exam 1 possible before 14 Feb 2026 – 10:30 h
  - ⇒ withdrawing from written exam 2 possible before 06 Apr 2026 – 07:30 h
- The cancellation deadline applies regardless of how you registered.
- If you registered via MTS on Moses ⇒ cancel via MTS on Moses. Double-check that you are no longer registered.
- If you registered via the examination office
  ⇒ cancel with the examination office and confirm your cancellation with Ms Groiß.
- Non-TU students that couldn't register via MOSES cancel their registration by sending an email to Ms Groiß (camilla.groiss@tu-berlin.de ✉).

[2] Check the latest procedures on the website of the Prüfungsamt

[3] Communicate the Attest to the Prüfungsamt, **not** to us.

Written exam: In-presence, on-site

The written exams will be closed-book, *pen & paper* exams.

Next steps

Everyone:

- Enroll into our ISIS course $\longrightarrow$
- Self-study: Maths primer + programming basics and visualization sheet

BCCN students:

- Assemble into groups of 2 or 3.
- Download the group template file on ISIS.
- Fill it out and send it to the tutors.
- Solve the first exercise sheet (Maths Primer) until the 29. Oct. 2025and send it to tutors.

To our ISIS course:

https://isis.tu-berlin.de/course/view.php?id=39708

Questions?

Post to the ISIS discussion forum.